Northern Devon Healthcare NHS Trust

Document Control

Title	
Protocol for Anaphylaxis in MIUs	
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1.1	October 2015	Final	Reviewed and amended Karen Watts / Fionn Bellis Approved at DTC September 2018
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Superseded Documents Protocol for Acute Anaphylaxis for use by Healthcare Providers in MIU's and WIC's	

Issue Date	Review Date	Review Cycle
September 2018	September 2021	Three years

Consulted with the following stakeholders:

- Fionn Bellis ED Consultant
- Michelle Cobby Outreach and Resuscitation Service Manager
- MIU Leads

Approval and Review Process • Lead Clinician for Emergency Department	
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1. Purpose

1.1. This protocol is for the use by nursing and paramedic practitioners employed by Northern Devon Healthcare Trust

2. Presenting Symptoms

2.1. Anaphylaxis is likely when all of the following 3 criteria are met:

- Sudden onset and rapid progression of symptoms
- Life threatening Airway and / or Breathing and / or Circulation problems
- Skin and / or mucosal changes (flushing, urticarial, angioedema)

The following supports the diagnosis

- Exposure to a known allergen for the patient

Remember:

- Skin or mucosal changes alone are not a sign of an anaphylactic reaction
- Skin and mucosal changes can be subtle or absent in up to 20% or reactions (some patients can have only a decrease in blood pressure, i.e. a Circulation problem)
- There can also be gastrointestinal symptoms (e.g. vomiting, abdominal pain, incontinence)

3. Key Points

3.1. Treatment of an anaphylactic reaction should be based on general life support principles:

- Use the Airway, Breathing, Circulation, Disability, Exposure (ABCDE*) approach to recognise and treat problems)
- Call for help early
- Treat the greatest threat to life first
- Initial treatments should not be delayed by the lack of a complete history or definite diagnosis

Patients having an anaphylactic reaction in any setting should expect the following as a minimum:

- Recognition that they are seriously unwell
- An early call for help
- Initial assessment and treatments based on an ABCDE* approach
- Adrenaline therapy if indicated
- Clinical Examination

3.2. Sudden onset and rapid progression of symptoms

- The patient will feel and look unwell.
- Most reactions occur over several minutes. Rarely, reactions may be slower in onset.
- The time of onset of an anaphylactic reaction depends on the type of trigger. An intravenous trigger will cause a more rapid onset of reaction than stings which, in turn, tend to cause a more rapid onset that orally ingested triggers.
- The patient is usually anxious and can experience a "sense of impending doom".

3.3. Life-threatening Airway and / or Breathing and / or Circulation problems

Patients can have either an A or B or C problem or any combination. Use the ABCDE approach to recognise these.

Airway problems:

- Airway swelling, e.g. throat and tongue swelling (pharyngeal / laryngeal oedema). The patient has difficulty in breathing and swallowing and feels that the throat is closing up.
- Hoarse voice.
- Stridor – this is a high-pitched inspiratory noise caused by upper airway obstruction.

Breathing problems:

- Shortness of breath – increased respiratory rate
- Wheeze
- Patient becoming tired
- Confusion caused by hypoxia
- Cyanosis (appears blue) – this is usually a late sign
- Respiratory arrest

Circulation problems:

- Signs of shock – pale, clammy
- Increased pulse rate (tachycardia)
- Low blood pressure (hypotension) – feeling faint (dizziness), collapse
- Decreased conscious level or loss of consciousness
- Anaphylaxis can cause myocardial ischaemia and electrocardiograph (ECG) changes even in individuals with normal coronary arteries
- Cardiac arrest

Circulation problems (often referred to as anaphylactic shock) can be caused by direct myocardial depression, vasodilation and capillary leak, and loss of fluid from the circulation. Bradycardia is usually a late feature, often preceding cardiac arrest.

The circulatory effects do not respond, or respond only transiently, to simple measures such as lying the patient down and raising the legs. Patients with anaphylaxis can deteriorate if made to sit up or stand up.

Disability:

The above Airway, Breathing and Circulation problems can all alter the patient's neurological status (**Disability problems**) because of decreased brain perfusion. There may be confusion, agitation and loss of consciousness.

Patients can also have gastro-intestinal symptoms (abdominal pain, incontinence, vomiting).

Exposure:

These should be assessed as part of the Exposure when using the ABCDE approach.

- They are often the first feature and present in over 80% of anaphylactic reactions
- They can be subtle or dramatic
- There may be just skin, just mucosal or both skin and mucosal changes
- There may be erythema – a patchy, or generalised, red rash
- There may be urticarial (also called hives, nettle rash, weals or welts), which can appear anywhere on the body. The weals may be pale, pink or red and may look like nettle stings. They can be different shapes and sizes, and are often surrounded by a red flare. They are usually itchy.

4. Treatment

Anaphylactic reaction?

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Airway, Breathing, Circulation, Disability, Exposure

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Diagnosis – look for:

- Acute onset of illness
- Life-threatening Airway and/or Breathing and/or Circulation problems[1]
- And usually skin changes

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Call for help

- Dial 999
- Lie patient flat
- Raise patient's legs

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Adrenaline[2]

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When skills and equipment available:

- Establish airway
- High flow oxygen
- IV fluid challenge[3]
- Chlorphenamine[4]
- Hydrocortisone[5]

Monitor:

- Pulse oximetry
- ECG
- Blood pressure

1 Life-threatening problems:
Airway: swelling, hoarseness, stridor
Breathing: rapid breathing, wheeze, fatigue, cyanosis, SpO_2 <92%, confusion
Circulation: pale, clammy, low blood pressure, faintness, drowsy/coma

2 Adrenaline: *(give IM)* IM doses of 1:1000 adrenaline (repeat after 5 min if no better)

- Adult: 500 micrograms IM (0.5 mL)
- Child more than 12 years: 500 micrograms IM (0.5 mL)
- Child 6-12 years: 300 micrograms IM (0.3 mL)
- Child less than 6 years: 150 micrograms IM (0.15 mL)

Adrenaline IV to be given **only by experienced specialists**
Titrate: Adults 50 micrograms; Children 1 microgram/kg

3 IV fluid challenge:
Adult – 500 – 1000mL
Child – crystalloid 20 mL/KG

Stop IV colloid if this might be the cause of anaphylaxis

	4 Chlorphenamine (IM or slow IV)	**5 Hydrocortisone** (IM or slow IV)
Adult or child more than 12 years	10mg	200mg
Child 6 – 12 years	5mg	100mg
Child 6 months to 6 years	2.5mg	50mg
Child less than 6 months	250 micrograms/kg	25mg

NB: Adrenaline may be given repeatedly every 5 minutes if required.(Resuscitation Council. UK)

5. Discharge Pathway

5.1. Patients who have received adrenaline should be transferred to secondary care for monitoring.

5.2. DOCUMENTATION TO BE COMPLETED

Clinical treatment record as per Documentation and record keeping policies. Copy of clinical treatment record to General Practitioner; to be sent to surgery as per Record keeping policy.

For patients being transferred to secondary care, ensure a copy of the clinical treatment record is sent with patient. A copy will also be sent to surgery in the normal manner.

5.3. BEFORE DISCHARGE ENSURE:

Those patients who have been referred for further acute intervention has appropriate transport to meet their needs, all relevant treatment has been prescribed and administered and correct information and documentation is given to the patient.

- The patient understands that if condition deteriorates or they have further concerns they should seek further advice.
- The patient demonstrates understanding of advice given during consultation.
- The patient has been provided with written advice leaflet to re-enforce advice given during consultation if available.
- The patient demonstrates an understanding of how to manage subsequent problems.

6. References

Consent Policy V3.3 (2014) NDHCT

Injectable Medicines Policy V1.1 (2013)NDHCT

Medicines Policy V1.0 (2015) NDHCT

Patient Group Direction Policy (2013) NDHCT

Resuscitation Council (UK) The Emergency Treatment of Anaphylactic Reactions (2010) (Reviewed 2017 with annotations for Emergency Departments not MIUs)

NICE Clinical Guideline 134, December 2011

APPENDIX A – Essential Documentation for All Patients Attending Unit or Centre

Adults Consent

Gain consent to be seen by a nurse practitioner

Gain consent for treatment and sharing information and document.

Clinical Presentation

If unwell assess for:

- Airway
- Breathing
- Circulation
- Disability
- Exposure
- Once treatment has been commenced:-

Document a full set of observations including neurological observations including Glasgow coma score if applicable.

Record EWS: if 7 or above arrange immediate transfer to secondary care.

Document pain score using numeric rating scale. For cognitively impaired patients document any signs of pain (e.g. grimaces or distress).

Safeguarding:

Ask the domestic abuse question, 'do you feel safe at home?'

- Assess for mental capacity and if person is a vulnerable adult.
- Assess for learning disability and whether patient has a hospital passport in place.
- Assess falls risk. Complete falls referral if applicable.

APPENDIX B – Essential Documentation for All Patients Attending Unit or Centre

Child and Young Persons under 18 Years Old Consent

Gain consent to be seen by a nurse practitioner

Gain consent for treatment and sharing information

Assess and document competency according to Fraser guideline if applicable.

Document the name of person(s) accompanying patient.

Clinical Presentation

If unwell assess for:

- Airway
- Breathing
- Circulation
- Disability
- Exposure

Record PEWS: if any one parameter is triggered transfer to secondary care or seek advice from medical practitioner.

Use guideline Traffic Light System (NICE) 2013 if applicable.

Use guideline Feverish Illness (NICE) 2013 if applicable.

Document pain score using FLACC, Wong Baker Faces or numeric rating scale.

Safeguarding

Complete safeguarding childrens questions (NICE 2003)

Any injury or bruise in a non-mobile infant and child: follow Safeguarding Children Policy (2018). These children must be reviewed by a Consultant in Emergency Medicine or a Consultant Paediatrician and a MASH (multi agency safeguarding hub) enquiry must be made.

DOCUMENT ALL FINDINGS IN THE CLINICAL TREATMENT RECORD

APPENDIX C – Training Competency Form

Protocol for the Management of Anaphylaxis

Protocol operational from October 2015 and expires end of October 2018

The registered health professional named below, being employees of Northern Devon Healthcare Trust based at ……. have received training and are competent to operate under this Standard Operating procedure

NAME *(please print)*	PROFESSIONAL TITLE	SIGNATURE	AUTHORISING MANAGER *(please print)*	MANAGER'S SIGNATURE	DATE

Keep original with the authorising manager and send a copy to: Karen Watts, Emergency Department, Northern Devon Healthcare Trust NHS, Raleigh Park, Barnstaple, Devon EX31 4JB